

Question 1 (3 points) ✓ *Saved*

LVQ is used to classify 4 input vectors X_k and their respective classes T_k for $k = 1$ to 4 into 2 classes such that $T_1=T_2=C_1=1$ and $T_3=T_4=C_2=2$. Let the 2 initial weights for classes C_1 and C_2 be selected as $W_1=X_2$ and $W_2=X_3$. Assuming that (i) all the 4 input vectors are to be presented orderly as X_1, X_2, X_3 , and X_4 in each iteration, (ii) an immediate update is applied to each weight vector within each iteration, and (iii) updates are carried out in all the weight vectors with equal minimum distances from an input vector.

Given the initial gain value $g(1)$ is 0.5 and $g(t+1) = \{v\} g(t)$ for $t \geq 1$, show the step-by-step calculation for the value of the sum of all the differences between elements obtained by subtracting W_2 from W_1 as $(W_1 - W_2)$ at the end of the first iteration (using $g(1)$) in the Answer Book and submit Your Answer rounded to 2 decimal places to the rectangular box below for grading. Given $X_1 = [0 \ 0 \ 1]$; $X_2 = [0 \ 1 \ 1]$; $X_3 = [1 \ 1 \ 0]$; $X_4 = [1 \ 0 \ 0]$.

Your Answer:

Answer

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